

MedDevice.AI: An AI-Powered Medical Device Assistant Using Retrieval-Augmented Generation and Optical Character Recognition

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Abstract—Medical devices play a vital role in disease diagnosis, patient monitoring, rehabilitation, and home healthcare. However, users often encounter difficulties in identifying appropriate medical devices due to fragmented information sources, conventional keyword-based search systems, and the absence of intelligent healthcare assistance. Existing platforms generally lack conversational interfaces capable of understanding natural language queries, analyzing uploaded medical reports, and providing personalized device recommendations.

This paper presents MedDevice.AI, an intelligent web-based medical device assistant that integrates Machine Learning, Retrieval-Augmented Generation (RAG), Optical Character Recognition (OCR), Speech Recognition, and cloud-based technologies to provide accurate and context-aware medical device assistance. Cloud-based platforms such as Flask [1] and Supabase [2], while the RAG pipeline utilizes FAISS-based retrieval to access relevant information from a structured medical device knowledge base, improving the relevance and reliability of chatbot responses. Optical Character Recognition, implemented using the Tesseract OCR engine, extracts clinical information from uploaded medical reports to recommend suitable medical devices based on detected medical conditions.

The application is implemented using Python and the Flask framework with a responsive frontend developed using HTML, CSS, Bootstrap, and JavaScript. Supabase provides secure authentication, cloud database management, personalized dashboards, chat history, and integrated medical device ordering. Functional validation of the implemented modules confirms the successful integration of AI-driven chatbot interaction, OCR-based report analysis, Retrieval-Augmented Generation, and cloud services into a unified healthcare platform. The proposed system provides an efficient, scalable, and user-friendly solution for intelligent medical device information retrieval and recommendation while demonstrating the practical application of Artificial Intelligence technologies in modern healthcare information systems.

Index Terms—Artificial Intelligence, Machine Learning, Healthcare, Medical Devices, Retrieval-Augmented Generation, Optical Character Recognition, Speech Recognition, Flask, Supabase

I. INTRODUCTION

Medical devices have become indispensable components of modern healthcare systems, supporting disease diagnosis, physiological monitoring, rehabilitation, and preventive healthcare. The rapid adoption of portable diagnostic equipment has enabled patients and caregivers to perform routine healthcare activities outside hospitals, increasing the demand for reliable and easily accessible medical device information. However, users often encounter fragmented information distributed across product catalogues, technical manuals, and multiple websites, making it difficult to identify suitable devices efficiently.

Most existing healthcare information systems primarily rely on static webpages or keyword-based search mechanisms. Although such systems provide basic product information, they are unable to understand conversational queries, analyze supporting medical documents, or deliver personalized recommendations. Consequently, healthcare professionals, biomedical engineering students, caregivers, and patients frequently spend considerable time searching through multiple resources before identifying appropriate medical devices.

Recent advancements in Artificial Intelligence (AI), Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), and Optical Character Recognition (OCR) have significantly improved intelligent information retrieval systems. These technologies enable conversational agents capable of understanding user intent, retrieving contextual knowledge from external documents, analyzing uploaded reports, and generating accurate responses. Furthermore, cloud computing platforms simplify authentication, database management, and personalized service delivery while maintaining scalability and reliability.

Motivated by these developments, this work proposes **MedDevice.AI**, an intelligent web-based medical device assistant that combines Machine Learning, Retrieval-Augmented Generation, OCR, Speech Recognition, and cloud technologies into a unified healthcare platform. Unlike conventional

systems, the proposed application not only answers medical device-related questions but also analyzes uploaded medical reports, recommends appropriate medical devices, supports voice interaction, stores personalized chat history, and enables secure online device ordering.

The major contributions of this research are summarized as follows.

- Design and implementation of an intelligent web-based medical device assistant that integrates conversational AI, Retrieval-Augmented Generation (RAG), Optical Character Recognition (OCR), Speech Recognition, and cloud-based services within a unified platform.
- Development of a Machine Learning-based intent classification module using TF-IDF Vectorization and Logistic Regression to accurately understand user queries related to medical devices.
- Integration of Retrieval-Augmented Generation with a structured medical device knowledge base to generate context-aware and reliable responses while supporting future knowledge expansion without retraining the classification model.
- Development of an OCR-based medical report analysis module that extracts clinically relevant information from uploaded reports and recommends suitable medical devices based on detected medical conditions.
- Implementation of secure user authentication, personalized dashboards, chat history management, and online medical device ordering using the Supabase cloud platform.
- Development of a modular and scalable system architecture that supports both text-based and voice-based interaction while facilitating future integration of advanced AI models and healthcare services.

The remainder of this paper is organized as follows. Section II reviews related research in intelligent healthcare systems. Section III presents the proposed methodology and system architecture. Section IV discusses implementation details. Section V presents experimental results and performance analysis. Finally, Section VI concludes the paper and outlines future research directions.

II. LITERATURE REVIEW

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) have significantly transformed intelligent healthcare information systems. Researchers have proposed conversational agents, retrieval-based question answering systems, medical recommendation platforms, Optical Character Recognition (OCR) techniques, and cloud-based healthcare applications to improve healthcare accessibility, decision support, and user interaction.

Bickmore *et al.* [3] demonstrated that conversational healthcare agents can improve patient engagement by providing personalized guidance and continuous health-related assistance through natural language interaction. Their work highlighted

the effectiveness of dialogue-based healthcare support; however, it primarily focused on patient communication rather than intelligent medical device recommendation or healthcare equipment management [?].

Lewis *et al.* [4] introduced the Retrieval-Augmented Generation (RAG) framework, which combines information retrieval with neural language generation to improve factual consistency and reduce hallucinations in AI-generated responses. By retrieving relevant information from external knowledge repositories before response generation, RAG significantly enhances the reliability of conversational AI systems operating in knowledge-intensive domains such as healthcare [4].

Jurafsky and Martin [5] presented comprehensive research on Natural Language Processing, speech technologies, intent detection, and dialogue management. Their work provides the theoretical foundation for modern conversational AI systems and demonstrates the importance of accurate language understanding for intelligent chatbot applications [?].

Recent developments in Optical Character Recognition (OCR), particularly the Tesseract OCR engine, have enabled accurate extraction of textual information from medical reports, prescriptions, and laboratory documents. OCR-based document analysis reduces manual data entry and supports automated extraction of clinically relevant information for healthcare applications [6].

Cloud-based platforms such as Flask [1] and Supabase [2] have simplified the development of scalable healthcare applications by providing secure authentication, database management, RESTful communication, and cloud storage. These technologies enable rapid deployment of intelligent web-based healthcare systems while ensuring maintainability and scalability [1], [2].

Although substantial progress has been achieved in conversational AI, retrieval-based knowledge systems, OCR, and cloud computing, most existing solutions focus on individual functionalities. Few systems integrate Machine Learning-based intent classification, Retrieval-Augmented Generation, OCR-enabled medical report analysis, Speech Recognition, secure cloud authentication, personalized dashboards, chat history, and medical device ordering within a single healthcare platform. The proposed **MedDevice.AI** addresses this research gap by combining these technologies into a unified, intelligent medical device assistance system that supports information retrieval, report analysis, personalized recommendations, and online device ordering through an integrated web-based architecture.

III. PROPOSED METHODOLOGY

The proposed system is designed as an intelligent web-based medical device recommendation and information retrieval platform. The architecture integrates Machine Learning, Retrieval-Augmented Generation (RAG), Optical Character Recognition (OCR), Speech Recognition, and cloud database services to provide accurate, personalized, and context-aware assistance.

The overall workflow consists of six major stages:

- 1) User Authentication

- 2) Query Processing
- 3) Machine Learning Intent Classification
- 4) Retrieval-Augmented Knowledge Search
- 5) OCR-Based Medical Report Analysis
- 6) Recommendation and Device Ordering

The modular architecture enables each component to operate independently while exchanging information through the Flask backend, resulting in improved maintainability, scalability, and system performance.

A. System Architecture

Figure 1 illustrates the overall architecture of the proposed system.

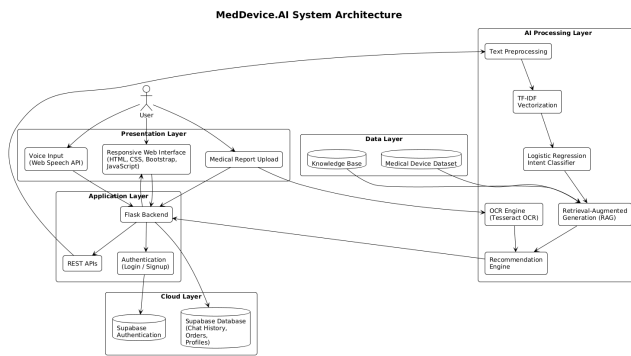


Fig. 1: Overall Architecture of MedDevice.AI

The architecture consists of four primary layers.

Presentation Layer: Provides an interactive web interface developed using HTML, CSS, Bootstrap, and JavaScript. Users can communicate through text or voice, upload medical reports, browse device information, and place medical equipment orders.

Application Layer: Implemented using the Flask framework, this layer manages routing, authentication, chatbot communication, OCR processing, recommendation generation, and API interactions.

Intelligence Layer: This layer integrates TF-IDF Vectorization, Logistic Regression, Retrieval-Augmented Generation (RAG), and Optical Character Recognition for intelligent understanding of user queries and medical reports.

Cloud Layer: Supabase provides authentication, cloud database services, chat history, user profiles, and secure medical device order management.

B. System Workflow

Figure 2 illustrates the operational workflow of the proposed platform.

Workflow of MedDevice.AI

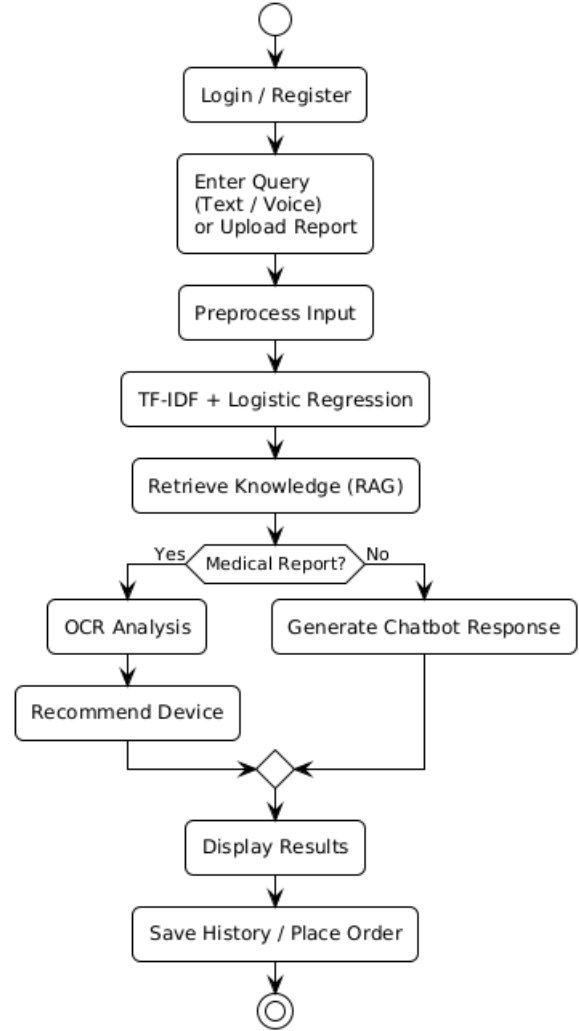


Fig. 2: Workflow of MedDevice.AI

The workflow proceeds as follows:

- 1) User logs into the system.
- 2) Query is submitted using text or voice.
- 3) Flask backend receives and preprocesses the input.
- 4) TF-IDF converts text into numerical feature vectors.
- 5) Logistic Regression predicts user intent.
- 6) Relevant knowledge is retrieved using Retrieval-Augmented Generation.
- 7) If a medical report is uploaded, OCR extracts clinical information.
- 8) The chatbot generates an appropriate response.
- 9) Recommended medical devices are displayed.
- 10) User may place an order using the integrated ordering module.
- 11) Chat history and order details are securely stored in Supabase.

C. Machine Learning Pipeline

The chatbot employs a supervised machine learning pipeline for intent classification. Initially, textual queries undergo pre-processing, including lowercase conversion, punctuation removal, stop-word elimination, and tokenization. The cleaned text is transformed into numerical feature vectors using the Term Frequency–Inverse Document Frequency (TF-IDF) technique [7].

The resulting vectors are provided as input to a Logistic Regression classifier trained using labeled medical device queries. The predicted intent is subsequently utilized to retrieve the most relevant medical device information from the structured knowledge base.

D. Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) [4] enhances chatbot performance by combining machine learning prediction with knowledge retrieval from curated medical datasets. Instead of relying solely on learned parameters, the chatbot retrieves contextually relevant device information before constructing the final response.

This retrieval strategy significantly improves factual accuracy, minimizes irrelevant responses, and enables continuous expansion of the knowledge base without retraining the classifier.

E. Optical Character Recognition

The OCR module allows users to upload laboratory reports, prescriptions, and medical documents. Text extracted using the Tesseract OCR engine [6] is analyzed to identify medical conditions, diagnostic keywords, and clinical observations.

These extracted terms are subsequently matched against the medical device knowledge base to recommend appropriate diagnostic or monitoring equipment, thereby providing personalized assistance beyond conventional chatbot interaction.

F. Mathematical Model

The TF-IDF representation [7] of a term is computed as

$$TFIDF(t, d) = TF(t, d) \times \log \left(\frac{N}{DF(t)} \right) \quad (1)$$

where

- $TF(t, d)$ denotes the frequency of term t in document d .
- $DF(t)$ denotes the number of documents containing the term.
- N represents the total number of documents.

The Logistic Regression classifier [8] estimates the probability of each intent using

$$P(y = 1|x) = \frac{1}{1 + e^{-(w^T x + b)}} \quad (2)$$

where

- x is the TF-IDF feature vector,
- w represents model weights,
- b denotes the bias parameter.

The predicted intent with the highest probability is selected for knowledge retrieval and response generation.

IV. SYSTEM IMPLEMENTATION

The proposed MedDevice.AI system was implemented using Python [9] and the Flask framework [1]. The application follows a modular architecture that separates the presentation layer, application logic, machine learning components, and cloud database services. This modular design improves maintainability, scalability, and future extensibility.

The frontend was developed using HTML5, CSS3, Bootstrap, and JavaScript to provide a responsive and user-friendly interface across desktop and mobile devices. The backend was implemented using Flask, which manages routing, user authentication, chatbot communication, OCR processing, machine learning inference, and interactions with the cloud database.

Medical device information is maintained within a structured dataset containing device categories, specifications, applications, manufacturers, accuracy, portability, price range, and related clinical information. User queries are preprocessed using tokenization and text normalization before being transformed into TF-IDF feature vectors. The Logistic Regression classifier predicts the query intent, while the Retrieval-Augmented Generation (RAG) module retrieves relevant contextual information from the knowledge repository before generating the final response.

Supabase [2] provides cloud-based authentication, user profile management, chat history storage, and medical device order management. The integration of cloud services enables secure data storage while simplifying deployment and future scalability.

A. Software and Hardware Requirements

Table I summarizes the development environment used for implementing the proposed system.

TABLE I: Software and Hardware Requirements

Component	Specification
Operating System	Windows 10/11 or Linux
Programming Language	Python 3.x
Backend Framework	Flask
Frontend	HTML, CSS, Bootstrap, JavaScript
Machine Learning	Scikit-learn
Database	Supabase PostgreSQL
OCR Engine	Tesseract OCR
Speech Recognition	Web Speech API
Development IDE	Visual Studio Code
Browser	Google Chrome / Microsoft Edge
RAM	Minimum 8 GB
Processor	Intel Core i5 or equivalent

B. Functional Modules

The MedDevice.AI platform consists of the following functional modules:

- 1) User Registration and Secure Authentication
- 2) AI Chatbot for Medical Device Queries
- 3) Retrieval-Augmented Knowledge Retrieval
- 4) OCR-Based Medical Report Analysis
- 5) Speech-Based Query Input
- 6) Personalized Dashboard
- 7) Chat History Management

- 8) Medical Device Ordering
- 9) Cloud Database Management

Each module operates independently while communicating through the Flask backend, thereby improving maintainability and simplifying future feature integration.

C. User Interface

The user interface was designed to provide a clean and intuitive interaction experience. Navigation is organized into dedicated pages for authentication, chatbot interaction, report analysis, dashboard access, and device ordering. Bootstrap components ensure responsive layouts, while JavaScript enables dynamic content updates without requiring full page reloads.

D. AI Chatbot Interface

The AI chatbot serves as the primary interaction module of MedDevice.AI, enabling users to obtain medical device information through natural language conversations. The chatbot combines TF-IDF Vectorization, Logistic Regression, and Retrieval-Augmented Generation (RAG) to understand user queries and generate context-aware responses.

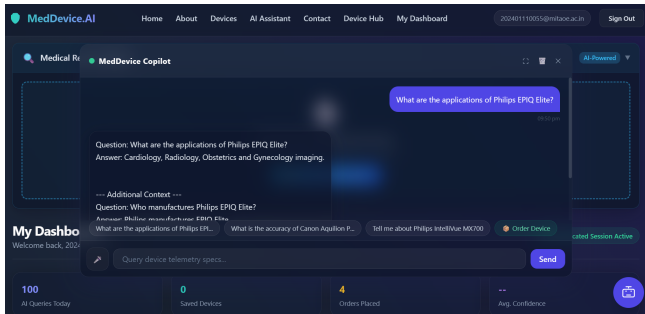


Fig. 3: AI Chatbot Interface

E. Medical Report Analyzer

The Medical Report Analyzer allows users to upload laboratory reports or prescriptions for OCR-based text extraction. The extracted clinical information is processed to recommend appropriate medical devices based on the detected medical conditions.

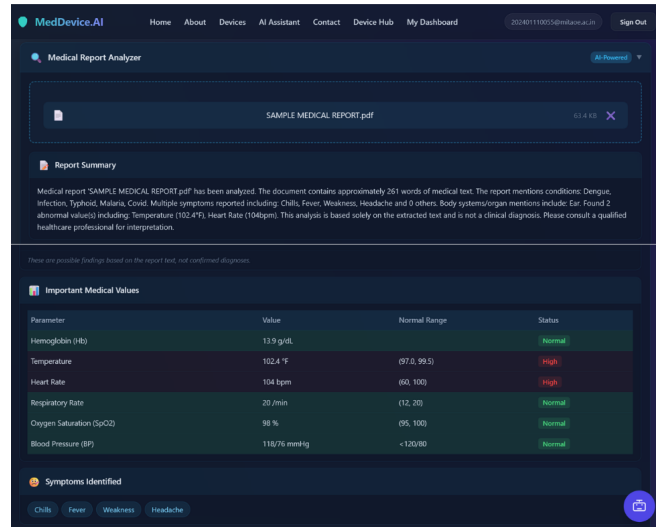


Fig. 4: OCR-Based Medical Report Analyzer

F. Medical Device Ordering

The ordering module enables authenticated users to submit medical device purchase requests directly through the application. Order details are securely stored in the cloud database, providing an integrated healthcare service within the platform.

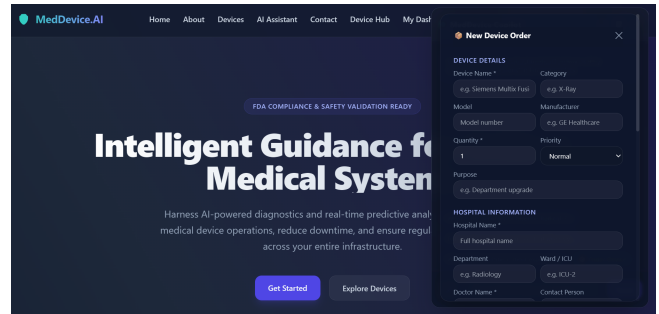


Fig. 5: Medical Device Ordering Module

V. RESULTS AND DISCUSSION

The proposed MedDevice.AI system was successfully implemented and functionally validated as an intelligent healthcare information platform integrating Machine Learning, Retrieval-Augmented Generation (RAG), Optical Character Recognition (OCR), Speech Recognition, and cloud-based services. Functional validation confirmed the correct operation of the major system modules, including secure user authentication, AI chatbot interaction, OCR-based medical report analysis, Retrieval-Augmented Generation, personalized medical device recommendation, and medical device ordering through the integrated cloud database.

Unlike conventional healthcare information systems that primarily rely on keyword-based searching, the proposed platform combines TF-IDF Vectorization, Logistic Regression, and FAISS-based Retrieval-Augmented Generation to provide context-aware responses to user queries. The retrieval mechanism enables the chatbot to access relevant information from

the medical device knowledge base without requiring retraining whenever new information is added, thereby improving system scalability and maintainability.

The OCR module, implemented using the Tesseract OCR engine, successfully extracts textual information from uploaded medical reports. The extracted clinical keywords are analyzed to identify relevant medical conditions, enabling the recommendation engine to suggest appropriate medical devices. Furthermore, the integration of Speech Recognition provides an alternative interaction method, improving accessibility for users who prefer voice-based communication.

Table II summarizes the functional validation of the major modules implemented in the proposed MedDevice.AI system.

TABLE II: Functional Validation of the Proposed System

System Module	Status
User Authentication	Successful
AI Chatbot Interaction	Successful
OCR-Based Report Analysis	Successful
RAG Knowledge Retrieval	Successful
Medical Device Recommendation	Successful
Speech Recognition	Successful
Medical Device Ordering	Successful
Supabase Database Integration	Successful

The functional validation demonstrates that the proposed system integrates multiple Artificial Intelligence technologies into a unified and user-friendly healthcare platform. The modular architecture improves scalability, maintainability, and extensibility, allowing future enhancements without major architectural modifications. Although the present work focuses on implementation and functional validation, future research will include comprehensive quantitative evaluation using metrics such as intent classification accuracy, response time, retrieval precision, and user satisfaction studies to further assess the performance of the proposed system.

VI. CONCLUSION AND FUTURE WORK

This paper presented **MedDevice.AI**, an intelligent web-based medical device assistant that integrates Machine Learning, Retrieval-Augmented Generation (RAG), Optical Character Recognition (OCR), Speech Recognition, and cloud computing technologies into a unified healthcare platform. The proposed system was developed to simplify medical device information retrieval by enabling users to interact through natural language, upload medical reports for automated analysis, receive context-aware device recommendations, and manage device orders through a secure cloud infrastructure.

The implementation demonstrates the practical integration of TF-IDF Vectorization and Logistic Regression for intent classification, Retrieval-Augmented Generation for knowledge retrieval, OCR for extracting information from medical reports, and Supabase for authentication and cloud database management. The modular architecture improves maintainability, scalability, and extensibility while providing an intuitive user experience through a responsive web interface.

Compared with traditional keyword-based healthcare information systems, the proposed platform combines conversational AI, retrieval-based knowledge access, report analysis, voice interaction, and personalized user services within a single application. These features improve accessibility and usability for patients, healthcare professionals, biomedical engineering students, and caregivers seeking reliable medical device information.

Although the current implementation successfully demonstrates the feasibility of the proposed architecture, several opportunities remain for future enhancement. Future work will focus on integrating Large Language Models (LLMs) for more advanced conversational capabilities, multilingual support, real-time medical device inventory management, image-based medical device identification, electronic health record integration, and deployment as a cross-platform mobile application. Comprehensive performance evaluation using larger datasets and user studies will also be conducted to further validate system effectiveness.

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